

Digitalization in European schools

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1 Introduction

This report aims to study the digitalization of European K12 education (from primary school to high school) by analyzing the existent and the impacts of certain measures on students' learning. We will provide trade-off insights on what is actually improved or not, and under which conditions. By digitalization, we mean the incorporation of Information and Communication Technologies, through for example online courses, computers as notebooks, online platforms and quizzes.

2 Current State of Digitalization in European Schools

The European Union produced in 2019 a very complete description of the technology available and used in schools [4]. It divides schools in 3 levels: primary school, middle school and high school. This report provides verified information, with standardized questions answered in all European countries. We will use the data from this report to assess the current hardware and connectivity situations in European countries. In the report, we will encounter the notions of ISCED 1, ISCED 2 and ISCED 3 which respectively correspond to primary, middle and high school.

2.1 Hardware

We start our analysis by examining the distribution of computers in primary, middle and high schools in Europe. In figure 1, we provide information from [4] on the average number of students per computer. There are in average 18 students per provided computer in primary schools, 7 for middle and high schools. However, it highly depends on the country with a standard deviation of 4.8, as shown in the figure below.

Fig.1.6.c: Number of students per computer (desktop computers, laptops, notebooks, tablets)
(ISCED 3, country and EU level, 2017-18)

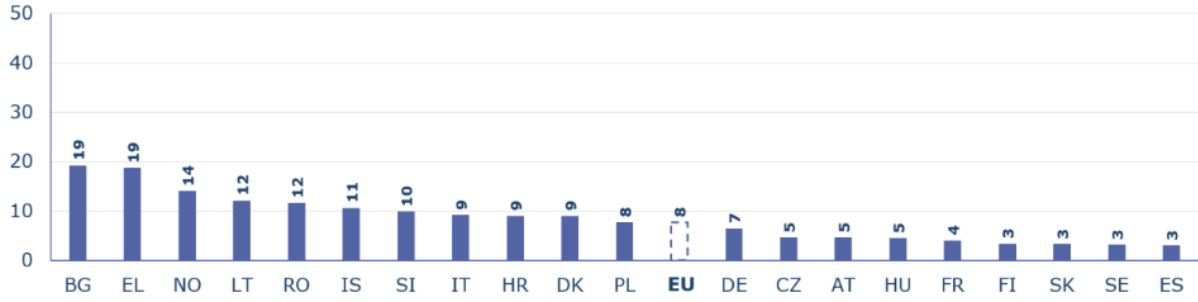


Figure 1: Number of students per computer grouped by country. Figure 1.6.c from [4]

Teachers surveyed in 2019 [4] considered the number of computers and tablets an obstacle for proper teaching in ICT. This statement alone could push decision-makers to invest in more computers to improve ICT teaching. However, this investment is not the only way to improve the teaching. The priority should be given to increasing the allotted time and teacher's competences. These factors are evaluated of equal importance by the teachers, and represent no significant environmental impact compared to new hardware (cf. section 3.5).

According to the report, a lot of hardware is broken in schools. A bit more than one third of students attend schools with less than 90% of functional equipment. No explicit analysis of the reasons behind this phenomenon were provided. We can identify several likely causes:

- small schools do not have an IT team, which could regularly test the hardware, identify the issues and take care of the repairing process.

- a lot of electronic devices are not repairable by design. Programmed obsolescence is a worsening phenomenon, which the Right To Repair movement is trying to counter. For example, the lifespan of smartphones decreased from 4.7 years in 2014 to 2.7 years in 2020. [28]

Repairing is of course preferable to buying new hardware in order to limit social and environmental impact.

States have leverage on the repairability of devices through legislation. Schools can act on the former cause, for example by designating someone responsible for regular hardware testing. Data could be collected at regional, national or even continental scale in order to measure how repairability is increased with such measures. This would help put in place a plan for digitalization including repairability.

2.2 Connectivity and Online tools

There are large inequalities of connectivity among schools of EU countries. Indeed, as we see on Figure 2.2, on average, less than 1 out of 5 of European students has access to high speed Internet (>100 mbps). This has probably a rather strong correlation with the development of the countries and its wealth, as we can observe on Figure 2.2 and Figure 2.2, the nations with the highest internet speed have higher GDP per capita.

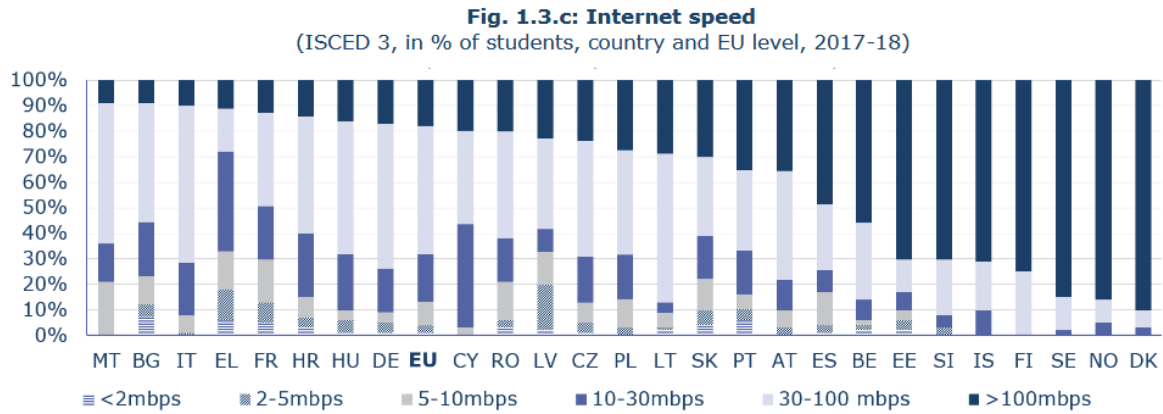


Figure 2: Share of ISCED 3 students having corresponding internet speed at school grouped by EU country. Figure 1.3.c from [4]

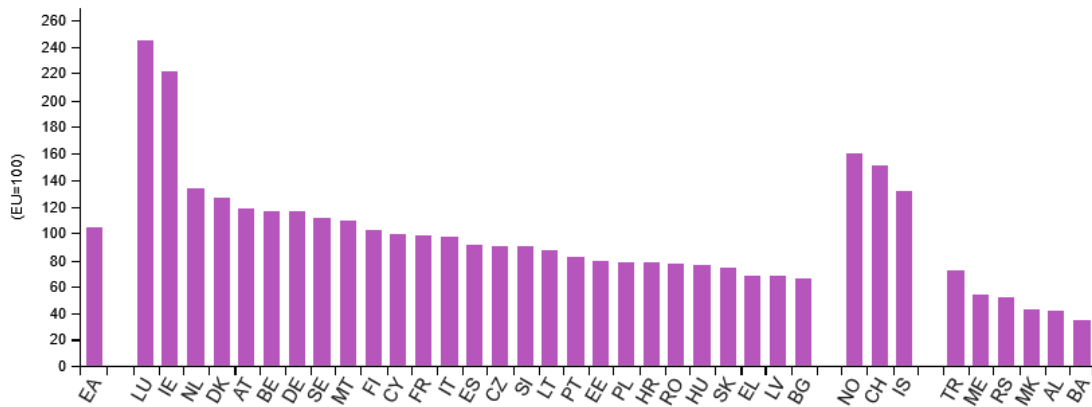


Figure 3: Index of GDP per capita in EU countries in 2023. Figure from ec.europa.eu

However, 65% of ISCED3 students have access to a Virtual Learning Environment (VLE) which consists of having an e-mail address, having access to online resources and attending courses that feature digital devices. Again, we can notice that the presence of a VLE in a school might be correlated to the wealth of its nation. A VLE enables students to work better outside of school, create working groups, eases the communication between teachers and students and can increase the personal feedback from a professor to their students.

Fig. 1.11.c: Schools with a virtual learning environment
(ISCED 3, in % of students, country and EU level, 2017-18)

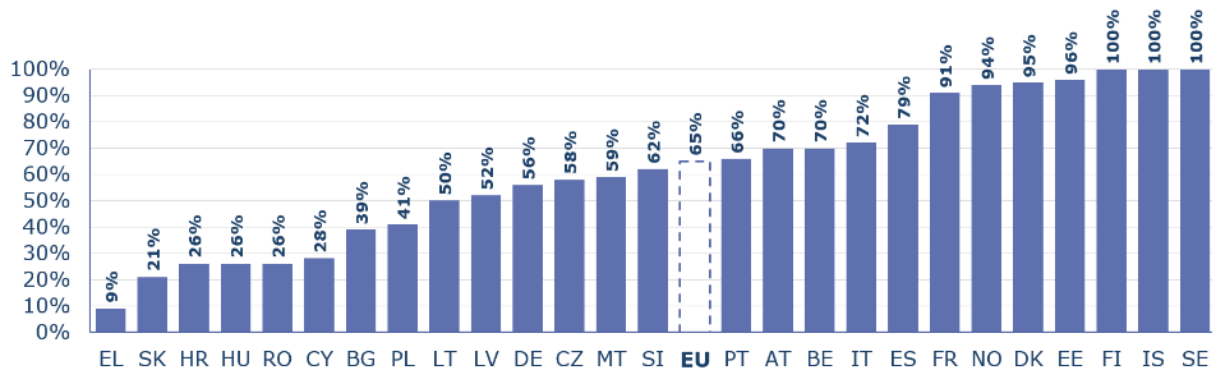


Figure 4: Share of ISCED 3 students studying in a school with a VLE grouped by country.
Figure 1.11.c from [4]

2.3 Student performances

There is no consensual metrics for the quality of learning. However, student performances can be measured via international assessments such as the PISA [1] framework. Quality of education is not the only factor, as living conditions and family education play also a big role. However, since these other factors are relatively constant, monitoring PISA evolution short-term can be a good metric for the impacts of new measures on the quality of learning. Currently, European schools obtain good scores, with only 6 countries below a 450/600 score. Estonia, Switzerland, the Netherlands, Ireland, United Kingdom, Poland, Belgium and Denmark are the best European countries. It does not make sense to compare directly their PISA scores with their digitalization state, because both are highly correlated to the GDP per inhabitant of the countries (ranked Monaco, Liechtenstein, Luxembourg, Ireland, Switzerland, Iceland, Norway, Denmark, Netherlands). As stated before, PISA change monitoring would be useful to assess the impact of measures in one country.

PISA however fails to measure the accessibility of education, as the sampling of students to sit the PISA test allows the exclusion of special schools or students with disabilities within schools.

2.4 Equality and inclusion

Equality and inclusion in European schools need improvement. For example, the European Court for Human Rights has dealt with six cases of discrimination in education for Roma children in Hungary, Czechia, Greece and Croatia, all after 2000, when a plan to reduce racial inequalities in schools had been adopted at the EU-level [25]. The French association "C.L.A.S.S.E.S" still testifies in 2025 that the schooling of Roma children and homeless children encounters many barriers. According to the association, the primary school system is not thought for inclusion.

Children and young people with disabilities, especially those with severe disabilities, are often denied access to education according to the Council of Europe's Commissioner for Human Rights. At the EU level, the proportion of students leaving school early is 9.5 percentage points higher for students with disabilities than for those without a disability [5].

2.5 Environmental impact

There is a lack of studies on the environmental impact of school digitalization. We strongly encourage that the European Union invests in such studies.

In section 3.5, we provide some estimation of the impact of the future digitalization, in particular regarding hardware.

To give magnitude orders, there are approximately 60 millions of students in K12 European schools, with an average of 8 students per computer, meaning 7.5 millions of computers. This is of course a wide approximation since the number of students per computer varies between countries. We mention in section 3.5 that a computer produces around 286 kg CO₂eq emissions throughout its lifetime, so all European education computers currently amount to 2 Megatons of CO₂eq emissions.

2.6 Limitations

Limitations are perceived both by teachers and students but teachers seem more likely to feel it. They express 3 types of limitations :

1. Equipment-related : Lack of equipment, usable equipment, poor connectivity, out of date equipment.
2. Pedagogy-related : Insufficient online content, difficulty of integration in the curriculum, lack of pedagogical support for teachers.
3. Attitude-related : Resistance of parents; resistance of teachers; lack of interest of teachers; no or unclear benefit to use ICT for teaching.

Recent studies tend to support identical limitations [3] [27], even though the technologies might be different. They are ways to reduce these limitations, and the solution varies with the type of limitation. To decrease pedagogy-related limitations, studies show that providing Technological Pedagogical Content Knowledge (TPACK) is mandatory so that teachers can use and adapt them for their courses. The attitude-related limitations are more subtle to diminish as new technologies often require to shift from traditional learning with many lectures and strong focus on student memorization. Digital tools are mostly used for flipped classroom, collective project and interactive lessons, which require many teachers to redesign courses [3]. Olivia Wohlfart and Ingo Wagner [27] explored teachers' integration of digital tools throughout COVID-19 pandemic and found out that technology acceptance is strongly linked with user motivation that is shaped with the perceived usefulness and the perceived ease of use that build the attitude toward using. They also observed that the use of technology affects, often positively, this motivation. They also emphasise on the crucial role of external environment such as opportunities, administrative support and infrastructure quality.

3 Impact of different digitalization measures in the classroom

In the previous section, we established the current state of digitalization in European schools, both regarding hardware and connectivity.

We will now focus on establishing how digitization is beneficial and what risks it presents. The ultimate goal is to provide guidelines for the future development of digitalization in education.

We will divide our analysis into Hardware and IoT in class, digital classrooms and online

classes and finally AI. Both equality and environmental impacts will be discussed.

3.1 Introducing computers and digital technologies

In class, the computers are mostly used for text and photo editing as well as Internet searches. The students are getting acquainted with the use of ICT but it does not help much for leaning in the other subjects. About 10% of students also regularly use learning apps for other subjects [4].

An OECD study from 2015 [26] found that the simple provision of ICT equipment is not sufficient, but that ICT is only linked to better student performance when, amongst others, computer software and Internet connections help to increase students study time and practice. The addition of technology is useless if not paired with some pedagogical improvements [16]. In fact, using computers only improves learning (in domains other than ICT) if they allow collective work and games. An online quizz in teams is the perfect example of a learning improvement via computers in class. Flipped classrooms are also proven more effective. The students read the course at home and practice with exercises in class. This is facilitated by students having a computer or tablet at home with access to an online platform [9].

However, the presence of digital devices in classrooms and especially personal devices can negatively impact learning processes of students. They become more distracted during classes, using these tools to browse on the Internet, chat with friends or even play games. Surveys reported that 75% of student use digital devices for non-class related activities during lectures and 40% admit multitasking in every class [24]. Moreover, a meta analysis raised evidence on the fact that an intensive use of digital devices leads to a drop in students scholar performances. PISA scores of students were compared to a survey that they filled about how often they conducted online activities. PISA scores increased with online activities up to a optimal use of several times a week, where it started to decrease [33]. Another study [32] performed on an international level and over decades, also found that performances declined with the arrival of personal devices for students. This was also related to students' mental health, especially their social relationships. Devices increased the number of students feeling lonely to 5 percentage points.

3.2 Digital classroom and online classes

A Digital Classroom is defined by using electronic devices or platforms such as social media, multimedia, and mobile phones to teach students. This general usage enables the teachers to introduce many systems and new pedagogical tools.

They can for example create a Virtual Classroom which is defined as a virtual environment where a teacher can interact with students in real-time, share his resources, deliver his lecture, assess students' learning, collect feedback, and reply to their queries. [26] A digital classroom can also enhance access to educational resources. Indeed, thanks to the Internet and the rising level of connectivity in Europe, access to educational resources anytime is now possible seven days a week. Moreover, Teachers can use feedback loops to provide personal feedback to students and groups on their work in order to create a rather quick cycle of autonomous work and guidelines for students.

Digitization of education could also help students get involved in their work. But some studies show that this effect is very limited. For instance, researchers observed the impact of digital games for mathematics in K12 schools on motivation, and found out that gamification of mathematics courses has in fact very few impact on motivation and that attention due to

novelty does not last long [7].

However, the creation of a virtual classroom could encourage students to skip lectures or classes. Additionally, online classes and a very virtual learning environment tend to favour better students and increase inequalities. Studies also reported that pupils from primary and secondary school learning twice as fast with a teacher in person rather than a remote teacher. The phenomenon of Screen Inferiority designates the lesser understanding of a text if read on a screen instead on paper. The effect will not diminish when students are more exposed to the technology. It can impact up to 20% of understanding on a text, depending on aggravating factors such as the need to scroll and a fixed screen [8]. It is hence important that students do not switch completely from printed handbooks to online versions on an online classroom. The schools can put in place some measures to reduce the weight of school bags such as lockers, sharing of books between students etc. while keeping printed material. Teachers should also avoid relying entirely on slides instead of written instructions or exercises.

3.3 AI in education

AI growth in the recent years has led to the question of using such a tool in education, its use and its impacts on learning processes of students.

The main use for AI in education is personalised learning. With this method, an AI model analyses students' data individually in order to find their strength and weakness, and propose educational material to reduce the gap with other students. For example, a personalised learning software has been used to teach maths and improve the results of students by 30% [15].

The use of AI, especially generative AI, also shapes the learning processes in education. Some studies raise the shift in students' mentality between learning and grading. As AI allow to bypass learning processes but still manage to deliver advanced work. It reinforces the importance of final grade over the learning outcome of every task, causing the students to learn and remember much less from their work while teachers struggle grading as rewarding the most complete work does not necessarily mean rewarding the student that learned and worked the most [17].

Some schools are using AI to grade the students arguing this can reduce teachers' bias and save time. It can be used to give student almost instant feedback on their assignments and help them improve their work. However, studies show that AI model gave three different grades for the same exam when given three times for 40% to 45% of students [12]. The three corrected exams were digital consisted of free answers in English about logistics. We have to notice that human grading of an exam can also vary depending on various factors. But we have to remember that grading is a teacher's crucial process in order to identify students' struggles, find gap in their courses and update their pedagogy to achieve the best learning outcome for their particular class.

But the use of AI in education raises several issues that cannot be omitted. Conversational AI have negative impact on long term cognitive skills and threaten critical thinking [13]. The institutions must guarantee privacy and data protection as lot of sensible data is collected by AI models. The models also have to be trained with unbiased data and be fair, which means not discriminate student based on their gender, ethnic or beliefs. Currently, generative AI models create gender discrimination as, in a professional recruitment, AI would recommend 72% of male candidates versus 48% female for technical roles [19], and AI models also enhance racial discriminations, for instance in mortgage applications [21]. Finally, the models have

to be accessible for every person with disabilities and be transparent about their system to teachers and students [15].

3.4 Inclusion and Inequalities

Concerning inequalities, the introduction of ICT in class can be helpful in many ways. For example, being introduced to computers already in K12 education with a regular usage diminished the fear for girls to pursue a career in ICT [6].

Computers and IoT can also allow disability aids. Currently, according to WHO, around 2 to 4% of students have significant learning difficulties [34]. For example, a student with dyslexia will more easily read and write on a computer with regular characters, an automatic corrector can make up for dysorthographia etc. A study [34] found that students with learning difficulties report fewer negative emotions compared to their peers in virtual learning environments. On the hardware part, Human-Computer Interaction (HCI) is a field of study which aims at creating computers adapted to everyone. Captors in IoT can help with the dimming of light for students on the Autistic Spectrum, describe colors for students with daltonism etc. Of course, such technology also costs more and sufficient funding has to be provided.

This reduction of inequalities is however conditioned by the inclusivity of the digital content and hardware themselves.

Moreover, the access should also be as equal as possible for all students. If some student work is to be conducted on computers, a student who has access to a computer at home will be advantaged. Currently, around 80-90% of students have this advantage in the EU, which makes it even more difficult for the remaining students. Moreover, even if computer reduces gender bias, this bias is ultimately still limited by societal expectations and representation.

3.5 Environmental impacts

The digitalisation of education has an impact on the environment on many levels. However, studies are lacking on the specific case of digitalising the education. Therefore, we will discuss the environmental consequence of ICTs and digitalisation of other instances.

A systematic literature review on the use of ICTs for sustainability shows that research is focused on efficiency, which means reducing the relative energy and resources used to produce a good or a service. But it highlights the gap in research on possible use of ICTs for Consistency and Sufficiency, which respectively consist of creating a circular economy and reducing the "absolute level of resource and energy demand by way of rethinking needs or changing consumption and production habits or patterns" [31]. They emphasise the need to research these three fields equally, adding that efficiency creates a great number of rebound effects that limit the positive initial intent.

While we do not yet have studies on Consistency and Sufficiency, we already know some substantial direct effects of producing new hardware that have to be taken into account when discussing the cons of digitalizing education. The production and disposal of the ICT hardware currently impacts very negatively the environment and quality of life of countries like Ghana (huge toxic e-waste disposals [29] [11]), China (water pollution due to the hardware production [20]), Democratic Republic of Congo (extracting of cobalt and other rare-earth

minerals under armed group and child labour [23]).

Via hardware and use, the ICT sector is already responsible to 5% of the world Greenhouse Gases emissions and digitalization of schools would imply more direct usage. Students would also graduate with digitalization as a part of their way of thinking and researching, which would speed up the digitalization of the whole society.

As an example, let us examine the impact of providing every Slovenian student with a computer for school. This example does not cover all of the infrastructure and hardware needed behind, and only encompasses one aspect of digitalization.

Let us consider here that solid laptops are bought each year to 15 000 students entering the ninth grade (the students would keep them until the last grade). Each computer will amount to 286.1 kg CO₂eq emissions throughout its lifetime. [10] 10 500 tons of materials would have to be extracted to produce these laptops. [22] However, one could argue that the students would have acquired a laptop on their own if the schools had not provided one. Some would have had the money to buy better ones while others not. In this way, the state providing this hardware reduces inequality between students. Yet if students are getting the computers themselves, they can make sustainable choices, i.e. getting the device that an older sibling who was going to throw it away, or buy second-hand. In fact, in Europe, 56% of laptops are bought second-hand [2]. The number is probably even higher in students because of their limited financial resources.

If the state wants a neutral carbon impact when furnishing computers, it needs to go with refurbished computers. It also needs to only give computers to students who don't already have one. Of course, this is a challenge because second-hand devices are not standardized and it would be difficult to provide the same quality to everyone.

4 Guidelines

The aim of this section is to provide guidelines for developing digitalisation in education in the more sustainable way. The following guidelines are general and directed at large scale educational instance such as government or ministry :

- Every step of digitalisation should be preceded by discussions including every stakeholder, especially teachers and students but also parents, schools and government. Additionally, schools should provide formations for teachers and students that have limited digital knowledge.
- Use digital devices that are repairable and easy to maintain. Try to repair broken devices first and try to buy refurbished devices if necessary.
- Research should be made to explore the existing gap in the use of ICTs not for efficiency but also on whether they can help create a circular economy and reduce the overall consumed resources.
- Focus the use of digital to learn about ICTs as it is less effective for other topics.
- The use of digital devices in primary and secondary school is not the most beneficial so its use can be limited to a common access such as an IT room or computer available in the library.

- High school students may need a personal device to do research at home. The school could provide such devices only for students that do not own a personal computer already.

Teachers have important leverage on applying the guidelines, in particular on the restricted use of ICTs for the digital learning purposes. As we shown earlier they should avoid relying only on slides, prioritize in-person classes and consider inverted classes with care as it can increase inequalities and impact students' mental health. They should also be fully aware of the data privacy risks and bias before using AI, and not use it for correction purposes as its consistency cannot be guaranteed.

Our findings suggest that students should prioritise reading on paper, put in place strategies to reduce distraction (especially with portable devices), limit their use of generative AI, and maintain social interactions as much as possible.

Conclusion

In conclusion, ICT in class is mainly useful to learn about the ICT domain itself, which is nowadays present in most jobs and important as an educational subject. However, it does not by itself improve the quality of learning in other domains. Some improvements are still seen with ICT usage on the short term because it facilitates certain ways of teaching that work best on students such as games, collective and active learning. ICT can moreover be useful to include students with special needs, by providing personalized content, allowing to study more easily in specific environments, or allowing an interface such as a vocal reader. Studies have also yet to be conducted large-scale with the introduction of AI.

However, when students own personal devices, these devices take up a bigger place in their lives than pedagogical games at school. They create distractions and cause loneliness, causing in the long-term a drop in academic performances.

Finally, the digitalization of the education sector would have a non-negligible environmental impact. While there is no specific study yet on the subject, new hardware (especially IoT) would have to be produced and the energy consumption would grow.

Throughout this paper, we have seen that the education digitalization impacts the SDG Goal of Reduced inequalities. If hardware and software are designed in an inclusive way, it can help disabled students. However, if the technology is financially difficult to access and not designed in an inclusive way, it also has the potential to increase inequalities.

The goal of Quality Education is poorly influenced by the digitalization in all domains except for ICT itself. However, certain uses of ICT have showed the importance of including games and collective work for students.

As discussed in the Environmental impacts section, the production of hardware and increase of energy usage implied by the digitalization would have a negative impact on the goals :

- Clean Water and Sanitation in other parts of the world.
- Life on Land
- Life below water
- Climate action
- Sustainable cities and communities
- Responsible consumption and production
- Good health and well-being
- Peace, justice and strong institutions.

Some compromises have to be made to take into account all of the goals. Our guidelines aim for the best course of action to take all of the SDGs into account [14] [18].

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Annex: Methodology

In order to write our subject proposal, we read the conclusions of a lot of papers, especially from this literature review. [30] This allowed us to find some documented measures we wanted to investigate and divide our work, as written in the proposal.

For our first milestone, we read the EU documents on European schools' state of digitalization. We both read the reports, then Eléonore wrote about Hardware and Repairability while Adrien wrote about Connectivity and Online Tools.

This first milestone passed, we focused on the four measures we meant to assess:

– Introducing IoT in class and giving personal devices to students (Eléonore) – Conducting classes online (Eléonore) – Creating a Digital classroom and giving students access to online resources (Adrien) – Introducing latest technologies in the classroom such as AI, blockchain or robots (Adrien) When reading the papers entirely, and a lot of new papers, we realised how difficult it was going to be to separate the effects of these measures. We had a problem of identifiability. We started writing our conclusions altogether. Then, Adrien organized them in different categories. We ended up with two big categories instead of four: "Introducing computers and digital technologies", and "Creating a digital classroom and leading classes online". It made sense because this is how the measures were most often paired in reality, so their effects were measured together. Eléonore deepened the research on Inequalities, since they were not yet addressed by the general papers about digitalization in details.

When presenting our second project update, our teacher gave us some new insights and resources, and recommended that we now dive into the environmental impacts.

Adrien focused on the bigger picture: tackling the effects of digitalization in other fields and in general. Eléonore tried to find some data on the sector of education (it is non-existent) and did the simulation with computers.

After that, Adrien wrote some guidelines that we both agree on.

Our conclusion existed from the beginning and we added our most important findings along the way. We rephrased it a bit for fluidity but did not have to change it much. Eléonore identified all of the SDG goals that we had come across in our research.

Our next step is mainly the poster, and adapting to our teacher's next feedback.